



Semester 1 Examinations 2015

Exam Code(s) 1SD1, 1MDM1, 1IF1, 1MF1, 1MCR1, 1SD3
Exam(s) Higher Diploma in Software Design and Development
Higher Diploma in Software Design and Development,
Industry Stream
Masters in Information Technology
Masters in Software Design and Development
Masters in Digital Media
Masters Clinical Research

Module Code(s) CT511
Module(s) Database Development

Paper No.
Repeat Paper

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Instructions: Answer **questions 1 and 2** and 2 other questions
All questions carry equal marks

Duration 2 Hours
No. of Pages 6
Discipline(s) Information Technology
Course Co-ordinator(s)

Requirements:

MCQ
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper

Other Materials

Release to Library: Yes ☒ No ☐

PTO

Question 1

- a) Create an **EERD** for the following:
Bentley College comprises several schools: school of business, school of arts and sciences, school of education, and school of applied sciences. Bentley has no plans in the future to add additional schools. Each school (which is standalone) is administered by a Dean, however each Dean is assigned to one school only.

Each school operates several departments, with each department offering several courses. However, each course is offered by a particular department. A course consists of several sections. Each section is taught by a single professor. Whilst each professor is assigned to a particular department, each department employs several professors.

Each department is chaired by a departmental chair who is a professor, but not each professor has to be a chair of a department. Furthermore, each professor teaches more than one section of a course(s). A professor may also be on a research contract and teach no section of a course at all.

Each student may enroll in several sections (but not more than one per course per semester), and in each section there may be many students enrolled.

Each department teaches several students, however, each student belongs to (studies in) a single department. Each student has an advisor in his/her department. Each advisor advises several students. An advisor is also a professor, but not all professors advise students.

(15 marks)

- b) Define/explain each of the following terms (use examples where relevant):

Conceptual Data Modeling

Completeness Constraint

Disjointness Constraint

(6 marks)

- c) Compare and contrast the following terms:

Entity integrity and Referential integrity

Intranets and Extranets

(4 marks)

Question 2

CONTACT

ContactID	LastName	FirstName	Address
9990	Kelly	Ian	The Edge
9991	Kelly	Tom	The Side
9992	Smith	Susan	The Bottom
9993	Browne	Jeremiah	The Top

- (a) Given the table called CONTACT; write the SQL to :

- Insert the following values: 9994, smith, jane, 10 somewhere st.
- Update the FirstName to Richard where the ContactID is 9990
- Delete all records from the CONTACT table where the last name is Smith
- Count the number of records in the table CONTACT where the last name is Kelly and the ContactID is less than 9992

(12 marks)

(b) Given the following tables,

Branch

BranchNo	bCity
B003	Glasgow
B004	Bristol
B002	London

PropertyForRent

PropertyNo	pCity
PA14	Aberdeen
PL94	London
PG4	Glasgow

write the SQL which:

- Lists all properties and any branch offices that are in the same city
- Lists all branch offices and any properties that are in the same city
- Lists the branch offices and properties that are in the same city along with any unmatched properties – this includes rows that are unmatched in both tables

(9 marks)

(c) Explain the difference between a Determinant and a Functional Dependency. Use an example to illustrate your understanding.

(4 marks)

Question 3

(a) Bring the COURSE table below through 2NF. Ensure to go through each of the 5 steps. Ensure that you draw the Functional Dependencies Diagram.

(12 marks)

COURSE

CourseID	SemesterID	#Places	Course Name
IT101	2009-1	100	Programming
IT101	2009-2	100	Programming
IT102	2009-1	200	Databases
IT102	2010-1	150	Databases
IT103	2009-2	120	Web Design

(b) Explain why the SEMESTER table is not in 3NF. Bring it to 3NF. Map to relations and then draw the resultant tables.

(9 marks)

SEMESTER

Course	Semester	#Places	TeacherID	TeacherName
IT101	2009-1	100	332	Mr Jones
IT101	2009-2	100	332	Mr Jones
IT102	2009-1	200	495	Mr Bentley
IT102	2010-1	150	332	Mr Jones
IT103	2009-2	120	242	Mrs Smith

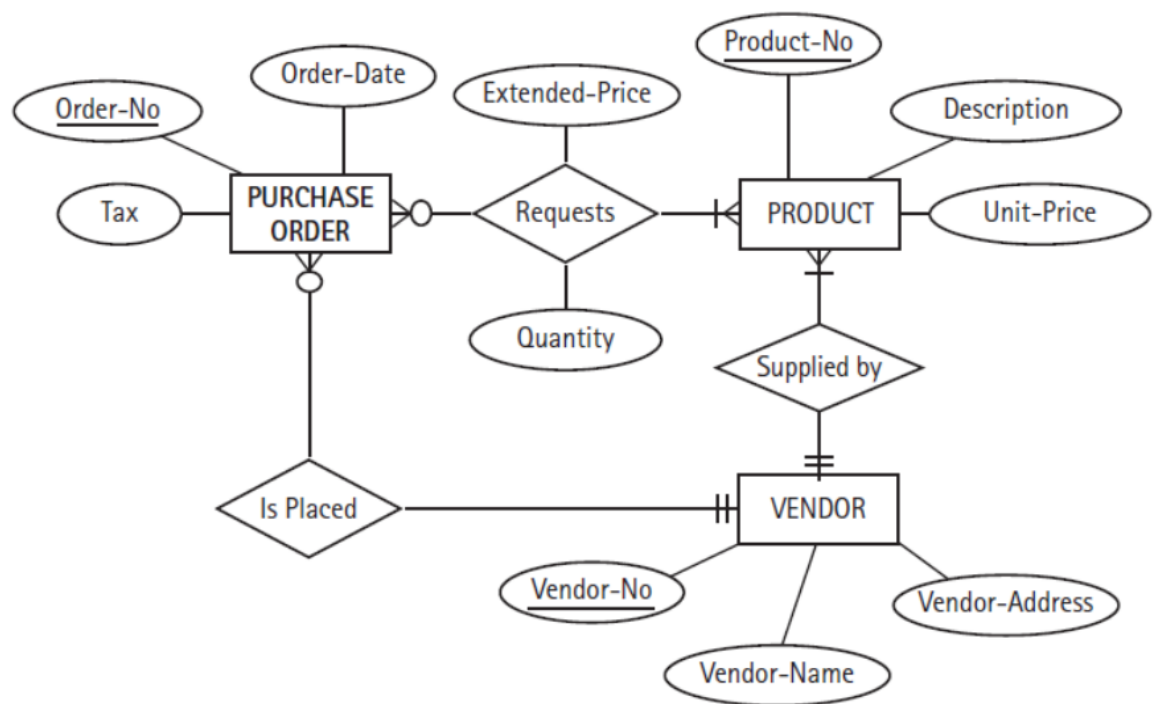
(c) Briefly explain the following terms – use examples to illustrate your answer:

- Data warehouse
- Foreign Key

(4 marks)

Question 4

a) Map the following to relations (use the short hand approach):



(10 marks)

b) Outline 4 limitations of the File Processing System approach. Use examples to illustrate your answer.

(8 marks)

c) Compare and contrast Logical Data Independence and Physical Data Independence (use a diagram to illustrate your answer)

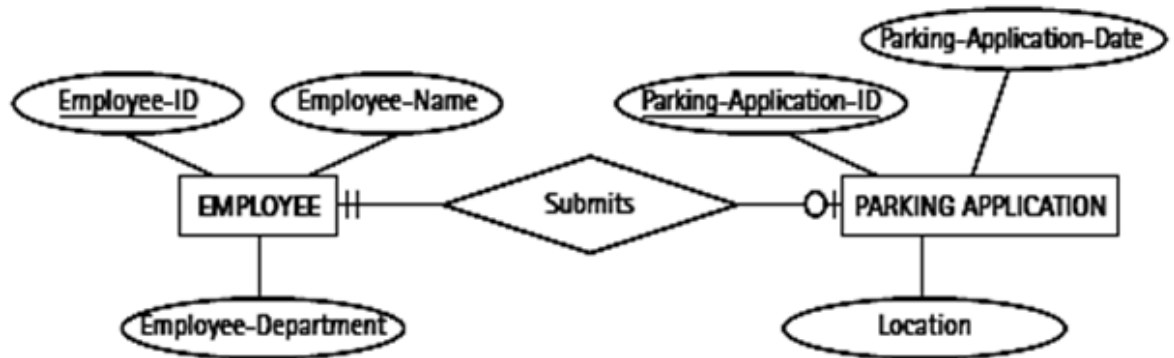
(7 marks)

Question 5

- a) Discuss why Denormalisation is carried out (include comments on benefits and drawbacks).

(5 Marks)

- b) Denormalise the following:



(7 Marks)

- c) Compare and contrast the roles of the Data and the Database Administrator

(8 Marks)

- d) Discuss what is meant by 'Big Data'

(5 Marks)

Question 6

- (a) Given the table called STAFF

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000.00	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000.00	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000.00	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000.00	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000.00	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000.00	B005

write the SQL so that for each branch having more than two members of staff, the branchNo (displayed as Branch Number), the number of staff in each branch (displayed as Number of Staff in the Branch) and the sum of their salaries (displayed as Sum of Salaries) is presented. The resultant table must be ordered by branchNo in reverse order (e.g. B007 before B003) (note: you will use aggregate functions).

(10 Marks)

b) Discuss what is meant by a Distributed Database Management System (in your discussion, include content on the advantages/disadvantages of a DDBMS)

(8 Marks)

c) Discuss how you would protect your database against virus attacks

(7 Marks)